AQA 3.2.2 and 3.2.3 Tests for ions (required practical 4)

AQA A-level Chemistry · Revision questions · Sheet 1 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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18 marks · about 25 minutes

Key information

- Group 2 cations: NaOH(aq) gives white $Mg(OH)_2$; H_2SO_4(aq) gives white $BaSO_4$ with Ba^{2+}
- NH_4^+: warm with NaOH(aq), damp red litmus turns blue; $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$
- CO_3^{2-}: dilute acid gives effervescence; CO_2 turns limewater cloudy
- SO_4^{2-}: add HCl-acidified $BaCl_2$(aq) — white precipitate; the acid removes CO_3^{2-} first
- Halides with HNO_3-acidified $AgNO_3$(aq): Cl^- white, Br^- cream, I^- yellow; F^- no precipitate

Section A: Short-answer questions

01 Barium metal is added to a large excess of water. Which observation is correct and complete?

[1 mark]

A a colourless solution
B a colourless solution with effervescence
C a dense white precipitate
D a dense white precipitate with effervescence

02 A precipitate formed in the silver nitrate test does not dissolve in concentrated ammonia solution. Identify the halide ion present.

[1 mark]

03 State what is observed when dilute aqueous sodium hydroxide is added to separate solutions of magnesium chloride and barium chloride.

[2 marks]

04 Give the ionic equation, including state symbols, for the formation of the precipitate in the test for sulfate ions.

[1 mark]

05 Describe a test to confirm that a solution contains hydroxide ions.

[1 mark]

06 Describe a test for ammonium ions in a solution. Give the observation for a positive result.

[2 marks]

Section B: Longer questions

07 This question is about Group 2 compounds.

07.1 Give the formula of the hydroxide of the element in Group 2, from Mg to Ba, that is least soluble in water.

[1 mark]

07.2 A student added 6 cm^3 of 0.25 mol dm^{-3} barium chloride solution to 8 cm^3 of 0.15 mol dm^{-3} sodium sulfate solution. The student filtered off the precipitate and collected the filtrate. Give an ionic equation for the formation of the precipitate. Show by calculation which reagent is in excess. Calculate the total volume of the other reagent which should be used by the student so that the filtrate contains only one solute.

[3 marks]

08 An ionic compound of a Group 2 element dissolves in water to form solution X. Separate samples of solution X are tested to identify the ions. The table shows the observations.

	Reagent added	Observation
Test 1	NaOH(aq)	White precipitate
Test 2	HNO_3(aq) and $AgNO_3$(aq)	Cream precipitate

08.1 Identify the Group 2 cation in solution X. Give an ionic equation for the reaction that occurs in Test 1.

[2 marks]

08.2 Identify the anion in solution X. Give an ionic equation, including state symbols, for the reaction that occurs in Test 2. Explain why $AgNO_3$(aq) is acidified.

[3 marks]

08.3 Give one property of barium sulfate that makes it useful in hospitals.

[1 mark]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**B**	*Source: 7405/3 Jun 2023 Q21*	1
02	Iodide / I^-	*Source: Practice question (written for this set)*	1
03	$MgCl_2$: (slight) white precipitate $BaCl_2$: no (visible) change / colourless solution	*Source: 7404/1 Nov 2020 Q05.3* Do not allow nothing / no observation	2
04	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$	*Source: Practice question (written for this set)*	1
05	Add universal indicator and it turns blue / purple **or** red litmus turns blue	*Source: 7404/1 Jun 2025 Q06.2* Allow add a named soluble magnesium salt — (slight) white precipitate	1
06	Add NaOH(aq) and warm damp red litmus paper (held at mouth of tube) turns blue	*Source: Practice question (written for this set)*	2
07.1	$Mg(OH)_2$	*Source: 7405/1 Jun 2018 Q04.4–04.5*	1
07.2	$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4$ $n(BaCl_2) = 1.5 \times 10^{-3}$ and $n(Na_2SO_4) = 1.2 \times 10^{-3}$ so $BaCl_2$ in excess 10 cm^3 (of 0.15 mol dm^{-3} sodium sulfate)	Working required for excess mark (or 3×10^{-4} mol excess $BaCl_2$)	3
08.1	M1 Mg^{2+} M2 $Mg^{2+} + 2OH^- \rightarrow Mg(OH)_2$	*Source: 7405/1 Jun 2025 Q01.4–01.6* M1 and M2 allow Ca^{2+}	2
08.2	M1 Br^- / bromide M2 $Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$ M3 to remove carbonate ions / other ions that would give a precipitate with silver nitrate		3
08.3	Insoluble **or** shows up on / blocks / absorbs X-rays	Ignore barium meal or X-rays unqualified	1